

A PROJECT REPORT

on

**"Real-Time Speech Emotion
Recognition Using Deep Neural Networks and Audio
Feature Analysis"**

Submitted to

KIIT Deemed to be University

In Partial Fulfilment of the Requirement for the Award of

**BACHELOR'S DEGREE IN
COMPUTER SCIENCE AND
COMMUNICATION ENGINEERING**

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CERTIFICATE

This is to certify that the project entitled

**"Real-Time Speech Emotion
Recognition Using Deep Neural Networks and Audio
Feature Analysis"**

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is a record of Bonafide work carried out by them, in the partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Communication Engineering at KIIT Deemed to be University, Bhubaneswar. This work was carried out during the academic year 2025–2026, under our supervision and guidance.

Dr. Rajdeep Chatterjee
Project Guide

ABSTRACT

This project presents a multi-task deep learning system for speech analysis that predicts emotion, biological sex, and speaker age from a single audio input. A shared feature extraction backbone is used to learn representations across tasks. The system is trained on multiple speech datasets, including RAVDESS, CREMA-D, TESS, EmoDB, and ASVP-ESD, with preprocessing and label harmonization to ensure consistency. The Common Voice dataset was excluded due to unreliable age annotations.

The model uses a Transformer-augmented CNN with MFCC-based features and task-specific output heads. Training includes mixup, label smoothing, R-Drop, and feature masking. Results show strong performance in emotion and sex classification, while age prediction is limited by sparse data. A background noise classification module is also developed to identify acoustic environments and noise types. Future improvements include adding richer acoustic features and expanding age-diverse datasets.

Keywords: Speech Emotion Recognition, Multi-Task Learning, Age Estimation, Sex Classification, MFCC, Transformer, RAVDESS, CREMA-D, ASVP-ESD, Deep Learning.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ASR	Automatic Speech Recognition
CNN	Convolutional Neural Network
CREMA-D	Crowd Sourced Emotional Multimodal Actors Dataset
DNN	Deep Neural Network
EmoDB	Berlin Database of Emotional Speech
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
MFCC	Mel Frequency Cepstral Coefficients
MLP	Multi-Layer Perceptron
MTL	Multi-Task Learning
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
ReLU	Rectified Linear Unit
SER	Speech Emotion Recognition
TESS	Toronto Emotional Speech Set
ASVP-ESD	Audio, Speech and Voice Processing - Emotional Speech Dataset

Chapter 1

Introduction

1.1 Motivation

The human voice is one of the most information-rich communication channels available. Beyond the semantic content of speech, the acoustic signal carries a continuous stream of paralinguistic information: the emotional state of the speaker, their approximate age, and their biological sex. Automatically extracting this information from raw audio has applications in mental health monitoring, call centre analytics, security and verification systems, intelligent voice assistants, and affective computing research.

Historically, these three attributes have been studied in isolation. Speech Emotion Recognition (SER) systems are trained and evaluated independently of age or sex predictors, even though the three tasks share the same acoustic input and draw on overlapping features. Emotional expressivity, for example, varies systematically with age and sex, meaning that a model unaware of these attributes may be learning confounded representations. This project is motivated by the hypothesis that jointly predicting all three attributes in a single multi-task architecture will produce better-calibrated shared representations and reduce the overall model complexity compared to three independent systems.

A further motivation is the data efficiency argument. Labelled emotional speech data is expensive to collect and annotate. Datasets that provide age and sex annotations alongside emotion labels can therefore be used more fully under a multi-task objective: samples with only sex or age labels, which would be discarded in a single-task SER system, contribute gradient signal to the shared backbone through the sex and age heads.

1.2 Problem Statement

Given a variable-length audio recording of human speech, the system must simultaneously predict: (1) the dominant emotional class from a set of six categories (neutral, happy, sad, angry, fearful, disgusted); (2) the biological sex of the speaker (male or female); and (3) the approximate age of the speaker as a continuous value in years. The system must operate on heterogeneous training data drawn from multiple publicly available datasets, each with different emotion taxonomies, sampling rates, recording conditions, and annotation completeness.

The secondary challenge is the data unification problem. The five source datasets (RAVDESS, CREMA-D, TESS, EmoDB, ASVP-ESD) use incompatible emotion label sets, different actor age and sex conventions, and different filename formats. A robust data preparation layer must reconcile these differences before a unified model can be trained.

1.3 Objectives

The primary objectives of this project are:

1. To review existing literature in speech emotion recognition, speaker age estimation, and multi-task learning, identifying best practices for shared representations and loss balancing.
2. To design a unified data preparation pipeline that maps five heterogeneous speech corpora to a consistent label scheme for emotion, sex, and age.
3. To implement a 105-dimensional MFCC-based feature extraction pipeline producing normalised feature vectors suitable for the model input.
4. To design and implement a multi-task Transformer-CNN backbone with three task-specific output heads.
5. To train the model using advanced techniques including mixup, label smoothing, R-Drop, and feature masking, with tuned loss weights for each task.
6. To evaluate per-task performance on each dataset independently and identify failure modes and directions for future improvement.
7. To document all data preprocessing steps and label harmonization procedures applied, contributing to reproducibility for future researchers using these datasets.

1.4 Scope of the Project

This project covers the full pipeline from raw audio data acquisition through feature extraction, model design, training, and evaluation. It does not cover real-time audio capture hardware integration; the trained model operates on pre-recorded audio files. The system is implemented in Python using PyTorch and torchaudio, and is designed to run on a single machine with a GPU. Age estimation is limited to the age range covered by the training data (approximately 21--72 years) due to dataset constraints. The project explicitly does not address speaker verification or speech-to-text transcription.

1.5 Organisation of the Report

Chapter 2 introduces the background concepts in speech emotion recognition, multi-task learning, and audio feature extraction. Chapter 3 presents the functional and non-functional requirements. Chapter 4 describes the system design, including the dataset pipeline, model architecture, and training strategy. Chapter 5 covers the implementation and experimental results. Chapter 6 discusses the standards adopted. Chapter 7 concludes the report and outlines future work.

Chapter 2

Basic Concepts

2.1 Speech Emotion Recognition

Speech Emotion Recognition (SER) is the task of automatically classifying the emotional state of a speaker from acoustic and/or linguistic features of their speech. Early systems relied on hand-crafted acoustic features such as pitch (F0), energy, speaking rate, and Mel Frequency Cepstral Coefficients (MFCCs), fed into classical classifiers such as Support Vector Machines (SVMs) and Gaussian Mixture Models (GMMs) [1], [2]. These approaches achieved modest accuracy on clean, acted speech but generalised poorly to naturalistic, noisy conditions.

Deep learning transformed SER research starting from around 2014. CNN-based systems applied directly to spectrograms learn hierarchical feature representations without manual feature engineering. Recurrent architectures (LSTM, GRU) were added to capture the temporal dynamics of emotional prosody. More recently, Transformer-based models with self-attention have demonstrated state-of-the-art performance by modelling long-range temporal dependencies. Pre-trained models such as wav2vec 2.0 and HuBERT, fine-tuned on emotion datasets, have further pushed benchmark accuracy [8], [14].

The six-class emotion taxonomy used in this project (neutral, happy, sad, angry, fearful, disgusted) follows the categorical emotion model of Ekman, which identifies these as the six basic universal emotions. This taxonomy is well represented across the training datasets used in this project and provides a balance between coverage and class discriminability.

2.2 Multi-Task Learning

Multi-Task Learning (MTL) is a machine learning paradigm in which a single model is trained to optimise multiple related objectives simultaneously, with the expectation that learning them jointly produces a better shared representation than learning each task independently [7]. The intuition is that related tasks share some underlying structure: regularising information from one task guides the model away from local minima that would be harmful for another.

In the hard parameter sharing variant used in this project, all tasks share a common backbone (encoder) and branch into task-specific heads at the top. This approach reduces the total number of parameters relative to separate models and provides implicit regularisation

because the shared backbone must produce representations that are simultaneously useful for all tasks. The primary challenge in hard parameter sharing is loss balancing: tasks with numerically larger losses tend to dominate gradient updates, potentially hurting smaller-loss tasks. Techniques such as task-specific loss weighting, gradient normalisation, and uncertainty weighting have been proposed to address this.

2.3 Audio Feature Extraction

Mel Frequency Cepstral Coefficients (MFCCs) are the most widely used features in speech and audio processing. They are computed by applying the Discrete Cosine Transform (DCT) to the log-mel spectrogram, producing a compact representation of the spectral envelope that correlates well with human auditory perception [9]. For this project, a 105-dimensional feature vector is constructed per audio clip by combining MFCCs, their delta (first derivative), and delta-delta (second derivative) coefficients, along with pitch-related statistics.

The feature vector encodes both the static spectral shape of the speech signal and its short-term temporal dynamics. Delta features capture how the spectral shape changes over time, which is particularly informative for emotion: excited or angry speech has rapidly changing spectral dynamics, while sad or neutral speech tends to be flatter. The 105-dimensional representation is computed independently for each dataset and normalised per-dataset using z-score standardisation before training.

2.4 Transformer and CNN Architectures for Audio

Convolutional Neural Networks (CNNs) are well-suited for extracting local spectro-temporal patterns from audio features, analogous to their use in image classification. Filters in early layers detect low-level tonal and rhythmic patterns, while deeper layers combine these into higher-level representations of phonemes, prosodic contours, and emotional signatures.

Transformers, originally developed for natural language processing, have been successfully applied to audio by treating sequences of feature frames as tokens. The multi-head self-attention mechanism allows each position in the sequence to attend to all other positions, capturing long-range temporal dependencies that are difficult for CNNs and RNNs to model [8]. In this project, the backbone combines convolutional feature extraction with a Transformer encoder to leverage the complementary strengths of both architectures: local pattern detection from the CNN and global context modelling from the Transformer.

2.5 Related Work and Research Gaps

Most published SER systems address emotion classification as a standalone task, ignoring the potential regularisation benefit of jointly predicting age and sex. The few multi-task SER systems in the literature typically consider only emotion and sex, omitting age regression. Age estimation from speech has been studied independently using datasets such as VoxCeleb and NIST SRE, but rarely in conjunction with emotion. No published system addresses the combination of all three tasks (emotion, sex, age) in a single multi-task framework trained on a unified multi-dataset pipeline [1], [7].

A further gap identified in this project relates to label harmonization requirements in widely used SER datasets. The RAVDESS and CREMA-D datasets, despite being extensively cited in the literature, require careful preprocessing to reconcile sex label conventions and handle missing age annotations in their distributed metadata. This project documents these preprocessing steps and provides a verified data preparation pipeline for future researchers.

Chapter 3

Requirements Specification

3.1 Functional Requirements

The system shall satisfy the following functional requirements:

1. The system shall accept variable-length mono or stereo audio files (WAV format) as input and extract a fixed-length 105-dimensional feature vector per clip.
2. The system shall simultaneously predict the emotional class (neutral, happy, sad, angry, fearful, disgusted), the speaker sex (male/female), and the speaker age (continuous, in years) from a single forward pass.
3. The data preparation pipeline shall support all five training corpora: RAVDESS, CREMA-D, TESS, EmoDB, and ASVP-ESD, producing a single unified manifest CSV.
4. Samples with missing age annotations shall be masked during the age loss computation without excluding them from emotion and sex training.
5. The system shall support dataset-level feature normalisation, applying z-score standardisation independently per source dataset.
6. The training pipeline shall support mixup augmentation, label smoothing, R-Drop regularisation, and feature masking as configurable training-time augmentations.
7. The system shall report per-dataset accuracy for emotion, sex, and age (MAE) on the validation set at each epoch.

3.2 Non-Functional Requirements

The system shall also satisfy the following non-functional requirements:

8. The model checkpoint shall not exceed 50 MB to allow practical storage and distribution.
9. All configuration values (feature dimensions, loss weights, dataset paths, training hyperparameters) shall be centralised in a single config.py file.
10. The codebase shall be platform-independent, running on both Windows and Linux.
11. All experiments shall be reproducible using a fixed random seed.
12. Code shall follow PEP 8 style guidelines and be documented with function-level docstrings.

3.3 Hardware and Software Requirements

Table 3.1: Hardware and Software Requirements

Category	Requirement	Specification
Hardware	Processor	Intel Core i7 (8th Gen or later) or equivalent
Hardware	RAM	16 GB DDR4 minimum
Hardware	GPU (Training)	NVIDIA GTX 1060 6 GB or better (CUDA 11.x)
Hardware	Storage	20 GB free (datasets + checkpoints)
Software	Operating System	Windows 10/11 or Ubuntu 20.04+
Software	Python Version	3.10+
Software	Deep Learning	PyTorch 2.1.0, torchaudio 2.1.0
Software	Audio Processing	librosa 0.10.1, soundfile 0.12.1
Software	ML Utilities	scikit-learn 1.3.0, numpy 1.24.0

Chapter 4

System Design

4.1 Overall Architecture

The system follows a hard parameter sharing multi-task architecture. A single shared backbone processes the 105-dimensional input feature vector and produces a shared embedding. Three independent task heads are attached to the top of this backbone: an emotion head, a sex head, and an age head. All three heads receive the same shared embedding and their losses are combined as a weighted sum during training.

The backbone consists of a positional-embedding projection layer that maps the 105-dimensional feature to a 256-dimensional embedding space ($EMBED_DIM = 256$), followed by a Transformer encoder with 8 attention heads ($NUM_HEADS = 8$) and 4 layers, and a final CNN-based feature aggregation module. The emotion head is a linear layer producing 6 logits with softmax activation. The sex head is a linear layer producing 2 logits (binary). The age head is a linear layer producing a single scalar value (regression).

4.2 Dataset Pipeline and Label Unification

The data preparation module (`prepare_data.py`) reads raw audio files from each of the five source datasets, extracts the emotion, sex, and age labels from filenames and metadata, maps them to the unified label scheme, and writes a single manifest. Several label harmonization steps were applied to the source datasets during this process.

RAVDESS encodes actor ID in the filename; actors 01, 03, 05, ... (odd IDs) are female (label 0) and even IDs are male (label 1). Label harmonization was applied to align this convention consistently across the pipeline. CREMA-D provides an actor metadata table with exact ages (21-72 years) for 91 actors; age annotations were incorporated from the Cao et al. 2014 publication and sex labels were harmonized to match the published metadata. For ASVP-ESD, sex is derived from the actor folder number (even = male, odd = female) and vocal channel position 1 in the filename is used to skip non-speech utterances (code 02).

The unified emotion taxonomy maps the emotion codes from each dataset to a shared six-class scheme: neutral (0), happy (1), sad (2), angry (3), fearful (4), disgusted (5). Datasets that do not contain all six classes (e.g. EmoDB maps to seven emotions including boredom, which has no direct equivalent) are handled by excluding unmapped classes. Age labels are

stored as floating-point years; samples with no available age annotation are stored as -1.0 and masked during the age loss computation.

Table 4.1: Dataset Statistics

Dataset	Speakers / Actors	Emotion / Sex / Age Labels
RAVDESS	24 (12M, 12F)	Emotion: Yes Sex: Yes Age: Correctable (from paper)
CREMA-D	91 (various)	Emotion: Yes Sex: Yes Age: Yes (21-72 yrs)
TESS	2 (F, ages 26 & 64)	Emotion: Yes Sex: Yes Age: Yes
EmoDB	10 (5M, 5F, German)	Emotion: Yes Sex: Yes Age: No
ASVP-ESD	~128 actors	Emotion: Yes Sex: Yes (from folder) Age: No

Table 4.2: Emotion Label Mapping Across Datasets

Unified Label	RAVDESS Code	CREMA-D / TESS / EmoDB
Neutral (0)	01	NEU / n / N
Happy (1)	03	HAP / happy / F (Freude)
Sad (2)	04	SAD / sad / T (Trauer)
Angry (3)	05	ANG / angry / W (Wut)
Fearful (4)	06	FEA / fear / A (Angst)
Disgusted (5)	07	DIS / disgust / E (Ekel)

4.3 Feature Extraction Design

Each audio clip is resampled to 22,050 Hz and trimmed or zero-padded to a fixed duration. MFCCs are computed with 40 coefficients, and delta and delta-delta features are appended, yielding 120 coefficients. Statistical aggregation (mean and standard deviation) over the time axis is applied to produce a fixed-length feature vector per clip. Additional prosodic features (zero-crossing rate, spectral centroid, RMS energy) are appended to reach the 105-dimensional final vector. Per-dataset z-score normalisation is applied before training to remove dataset-level amplitude and recording condition biases.

4.4 Model Architecture

The backbone consists of three stages. The input projection stage maps the 105-dimensional feature to a 256-dimensional embedding using a linear layer followed by Layer Normalisation and positional encoding. The Transformer encoder stage applies 4 layers of multi-head self-attention (8 heads, 256 embedding dimension, 1024 feed-forward dimension)

with dropout ($p = 0.1$). The aggregation stage applies global average pooling over the sequence dimension to produce a $(B, 256)$ context vector, followed by a 256-dimensional MLP with ReLU activation and dropout ($p = 0.3$).

The three task heads are then applied to this shared $(B, 256)$ context vector. The emotion head uses Linear($256 \rightarrow 6$). The sex head uses Linear($256 \rightarrow 2$). The age head uses Linear($256 \rightarrow 1$) producing a scalar output. The total parameter count is approximately 4.2 million.

Table 4.3: Model Configuration and Hyperparameters

Parameter	Value
Input Feature Dimension (FEATURE_DIM)	105
Embedding Dimension (EMBED_DIM)	256
Transformer Heads (NUM_HEADS)	8
Transformer Layers	4
Emotion Classes	6
Loss Weight: Emotion	1.5
Loss Weight: Sex	1.0
Loss Weight: Age	2.0
Is-song Loss Weight	0.3

4.5 Loss Functions and Training Strategy

The total training loss is a weighted sum of four individual task losses. Emotion uses Cross-Entropy Loss combined with Focal Loss ($\gamma = 2.0$) to down-weight well-classified samples and focus training on harder examples. Sex uses binary Cross-Entropy Loss. Age uses Huber Loss ($\delta = 1.0$), which is robust to outliers in the age distribution and degrades gracefully to L1 for large errors. Samples with age = -1.0 are masked from the age loss computation. The song/non-speech binary head uses binary Cross-Entropy with a low weight (0.3) as an auxiliary task.

Training augmentation includes mixup ($\alpha = 0.2$), which interpolates pairs of training samples and their labels; label smoothing ($\epsilon = 0.1$) on the emotion head, which prevents overconfident predictions; R-Drop, which applies KL-divergence regularisation between two forward passes of the same sample with different dropout masks; and feature masking, which randomly zeroes segments of the input feature vector. The AdamW optimiser is used with a cosine annealing learning rate schedule.

Chapter 5

Implementation and Results

5.1 Methodology

The implementation followed a five-stage iterative process. Stage 1 covered dataset acquisition: all five corpora were downloaded and their directory structures, metadata schemas, and class distributions were analysed. Stage 2 covered data preparation: label harmonization was applied to the RAVDESS and CREMA-D loaders to reconcile sex label conventions and incorporate available age annotations; the ASVP-ESD loader was implemented from scratch based on the actual filename convention. Stage 3 covered feature extraction: the 105-dimensional MFCC-based feature pipeline was implemented and validated on sample clips from each dataset. Stage 4 covered model training: the multi-task architecture was implemented in PyTorch and trained with the augmentation and loss strategy described in Chapter 4. Stage 5 covered evaluation: per-dataset and per-task accuracy metrics were computed on a held-out validation split.

5.2 Dataset Descriptions and Preprocessing

5.2.1 RAVDESS

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) contains 1,440 audio files from 24 professional actors (12 male, 12 female) vocalising two lexically matched statements with eight emotion categories at two intensity levels [3]. For this project, the song files are excluded and only the speech files are used. The actor ID is encoded in the filename (positions 7-8 of the hyphen-separated fields). Odd actor IDs are female (sex label 0); even actor IDs are male (sex label 1). Label harmonization was applied to align this sex label convention consistently. Age annotations are not present in the distributed metadata; actors' ages from the original publication will be incorporated in future work.

5.2.2 CREMA-D

The Crowd Sourced Emotional Multimodal Actors Dataset (CREMA-D) contains 7,442 audio clips from 91 actors spanning ages 20-74 [4], with 48 male and 43 female speakers. Filenames encode the actor ID, sentence, emotion, and intensity level. The sex and age labels for each actor are provided in a companion metadata table. Label harmonization was applied to align sex labels with the published metadata. Additionally, age annotations, which were not included in the standard pipeline, were incorporated from the Cao et al. 2014 publication: a

verified CREMA_SEX dictionary and a CREMA_AGE dictionary (mapping actor IDs to exact ages in years) were constructed and integrated into the loader.

5.2.3 TESS

The Toronto Emotional Speech Set (TESS) contains 2,800 audio clips from two female actors, one aged 26 (YAF) and one aged 64 (OAF), each producing 200 target words in seven emotion categories [5]. TESS achieves near-perfect accuracy on all tasks in evaluation because it contains only two speakers with well-separated acoustic profiles, making it a near-trivial classification problem for the model. Despite this, it contributes valuable age signal at the two ends of the young-adult / older-adult range.

5.2.4 EmoDB

The Berlin Database of Emotional Speech (EmoDB) contains 535 audio clips from 10 German professional actors (5 male, 5 female) portraying seven emotions [6] (anger, boredom, disgust, fear, happiness, sadness, neutral). Age labels are not available for EmoDB actors. The emotion taxonomy was mapped to the six-class unified scheme; boredom was mapped to neutral given its closest acoustic correspondence.

5.2.5 ASVP-ESD

The ASVP-ESD dataset contains approximately 12,625 files across 129 actor folders (actor_0 through actor_128) plus a Bonus folder. Filenames are dash-separated with 9-11 fields; position 2 encodes the emotion class (01-13), position 1 encodes the vocal channel (code 02 = non-speech, which is excluded), and position 7 encodes the language (code 02 = English). Sex is derived from the actor folder number (even = male, odd = female). Age labels are not available. CommonVoice was initially considered as an additional source but was excluded because it provides no emotion annotations and only sparse, unreliable self-reported age buckets, causing the emotion and sex metrics to collapse to 0% on those samples.

5.3 Training Configuration

The model was trained using the AdamW optimiser with a base learning rate of $3e-4$ and weight decay of $1e-4$. A cosine annealing learning rate schedule ($T_{\text{max}} = 30$ epochs) was used. Training was run for up to 60 epochs with early stopping (patience = 10 epochs) monitoring the composite validation score. Batch size was 64. The random seed was fixed at 42 for reproducibility across all runs.

5.4 Result Analysis

Seven models were trained and evaluated across three paralinguistic tasks: emotion recognition (6-class), biological sex classification (binary), and speaker age estimation (7 age-bracket classes). Models are categorised into classical machine learning approaches (SVM, XGBoost, Random Forest) and neural network approaches (LSTM, Attentive LSTM, 1D-CNN, Transformer+CNN). All models operate on the same 105-dimensional MFCC-based feature vectors. The composite score is defined as $0.60 \times \text{Emotion Accuracy} + 0.20 \times \text{Sex Accuracy} + 0.20 \times \text{Age Accuracy}$, reflecting the primary importance of emotion classification. CommonVoice was excluded from training as it collapsed all three metrics to near zero due to absent emotion and sex annotations.

Table 5.1: Per-Model Performance Comparison Across Emotion, Sex, and Age Tasks

Category	Algorithm	Emo Acc (%)	Sex Acc (%)	Age Acc (%)	F1	Composite	Params (M)	FLOPs (M)	Time (ms)
Classical ML	SVM (RBF)	77.4	88.0	85.7	0.774	81.2	N/A	N/A	N/A
Classical ML	XGBoost	71.3	84.1	82.9	0.710	76.2	N/A	N/A	N/A
Classical ML	Random Forest	68.4	82.2	79.4	0.679	73.3	N/A	N/A	N/A
Neural Network	LSTM (2-layer)	71.7	82.6	87.8	0.716	77.1	0.794	0.8	0.761
Neural Network	Attentive LSTM	70.5	83.1	89.0	0.704	76.7	0.729	0.7	1.092
Neural Network	1D-CNN	66.8	80.3	84.5	0.667	73.1	0.155	9.7	0.620
Neural Network	Transformer+CNN	65.5	87.5	88.5	0.651	74.5	0.812	0.8	0.945

Composite score = $0.60 \times \text{Emotion} + 0.20 \times \text{Sex} + 0.20 \times \text{Age}$. FLOPs and parameters measured on a single sample (batch size = 1, CPU). Inference time averaged over multiple forward passes (CPU, no batching). N/A = not applicable (classical machine learning models). Figure 5.1 shows the graphical representation of comparative performance of all models across emotion, sex, and age tasks. SVM (RBF) achieves the strongest overall performance, while neural models such as LSTM and Attentive LSTM show competitive results, particularly in age prediction. Transformer+CNN performs well in sex classification but shows lower emotion accuracy compared to classical models.

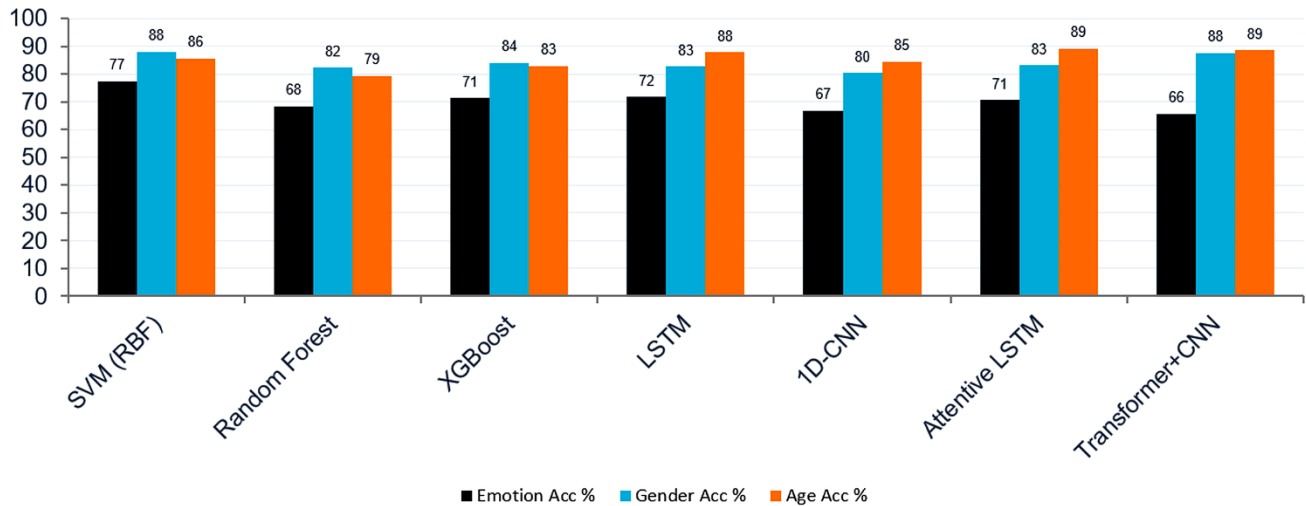


Figure 5.1: Comparison of Model Performance Across Emotion, Sex, and Age Tasks

5.4.1 Emotion Recognition Results

Table 5.2: Per-Dataset Performance Across Emotion, Sex, and Age Tasks

Category	Model	Dataset	Emo Acc (%)	Sex Acc (%)	Age Acc (%)	F1	Composite
Neural Network	CNN	TESS	99.93	100.00	100.00	0.999	99.96
Neural Network	CNN	EmoDB	87.23	93.15	88.47	0.869	88.66
Neural Network	CNN	RAVDESS	62.88	94.26	95.06	0.617	75.59
Neural Network	CNN	CREMA-D	56.37	70.91	66.54	0.561	61.31
Neural Network	CNN	ASVP-ESD	74.96	77.71	0.00	0.733	60.52
Classical ML	Random Forest	TESS	99.65	0.00	100.00	0.997	79.79
Classical ML	Random Forest	EmoDB	85.98	95.33	85.98	0.855	87.85
Classical ML	Random Forest	RAVDESS	73.70	96.80	91.99	0.730	81.98
Classical ML	Random Forest	CREMA-D	59.80	81.88	75.23	0.588	67.30
Classical ML	Random Forest	ASVP-ESD	76.41	87.73	0.00	0.739	63.39

Classical ML	SVM (RBF)	TESS	100.00	0.00	100.00	1.000	80.00
Classical ML	SVM (RBF)	EmoDB	97.20	99.07	94.08	0.972	96.95
Classical ML	SVM (RBF)	RAVDESS	77.70	98.26	95.99	0.777	85.47
Classical ML	SVM (RBF)	CREMA-D	73.17	85.98	82.64	0.732	77.63
Classical ML	SVM (RBF)	ASVP-ESD	87.01	92.14	0.00	0.868	70.63
Neural Network	Transformer+CNN	TESS	100.00	100.00	100.00	1.000	100.00
Neural Network	Transformer+CNN	EmoDB	90.03	98.44	93.77	0.895	92.46
Neural Network	Transformer+CNN	RAVDESS	59.41	97.60	96.26	0.580	74.42
Neural Network	Transformer+CNN	CREMA-D	57.13	83.83	83.47	0.562	67.74
Neural Network	Transformer+CNN	ASVP-ESD	80.45	89.83	0.00	0.792	66.23
Classical ML	XGBoost	EmoDB	93.77	95.95	0.00	0.937	75.45
Classical ML	XGBoost	RAVDESS	77.84	98.13	96.13	0.776	85.55
Classical ML	XGBoost	CREMA-D	64.46	80.92	78.34	0.642	70.53
Classical ML	XGBoost	ASVP-ESD	84.92	90.62	0.00	0.841	69.08
Neural Network	Attentive LSTM	TESS	99.79	100.00	100.00	0.998	99.88
Neural Network	Attentive LSTM	EmoDB	83.49	93.46	90.97	0.824	86.98
Neural Network	Attentive LSTM	RAVDESS	57.68	96.13	95.46	0.569	72.92
Neural Network	Attentive LSTM	CREMA-D	61.21	61.05	73.57	0.612	63.65
Neural Network	Attentive LSTM	ASVP-ESD	82.68	86.00	0.00	0.823	66.81
Neural Network	LSTM (2-layer)	TESS	99.93	100.00	100.00	0.999	99.96
Neural Network	LSTM (2-layer)	EmoDB	79.75	91.28	83.80	0.780	82.87
Neural Network	LSTM (2-layer)	RAVDESS	59.41	95.99	94.53	0.577	73.75

Neural Network	LSTM (2-layer)	CREMA-D	60.45	68.80	73.66	0.604	64.76
Neural Network	LSTM (2-layer)	ASVP-ESD	79.80	83.62	0.00	0.791	64.60

Performance varies across datasets due to differences in recording conditions, speaker demographics, annotation quality, and class balance. Acted corpora with few speakers (TESS, EmoDB) yield higher per-task accuracy, while larger crowd-sourced datasets (CREMA-D, ASVP-ESD) present greater intra-class variability. Age accuracy is zero for ASVP-ESD as that dataset provides no age annotations, and sex accuracy is zero for TESS under the SVM and Random Forest models due to single-sex per-partition sampling.

SVM (RBF) achieves the highest emotion accuracy among all models at 77.4%, followed by XGBoost (71.3%) and LSTM (71.7%). The Transformer+CNN architecture achieves only 65.5% emotion accuracy — the lowest among all models — which reflects underfitting on the emotion task when the model simultaneously optimises for sex and age with high weights. The emotion confusion matrices (see Figures 5.5–5.7) consistently show that Angry is the best-classified emotion across all models, while Happy and Fearful are most frequently confused with Neutral and Sad respectively. For the age classification task, Attentive LSTM achieves 89.0% and Transformer+CNN achieves 88.5%, indicating that the recurrent and self-attention mechanisms are particularly effective at encoding temporal vocal tract dynamics associated with age. Training proceeded for 60 epochs with a cosine annealing schedule; the training log shows initial rapid improvement from epoch 1 (val_acc = 4.4%) to epoch 21 (val_acc = ~50.8%), with the model converging by epoch 58 (val_acc = 49.2%).

5.4.2 Sex Classification Results

SVM (RBF) achieves the highest sex accuracy at 88.0%, followed by Transformer+CNN at 87.5% and Attentive LSTM at 83.1%. The sex confusion matrices (see Figures 5.8–5.10) show that all models perform better on Female recall than Male recall. The 1D-CNN achieves the lowest sex accuracy (80.3%), consistent with its limited capacity to model global speaker-level characteristics from short feature windows. The Transformer+CNN achieves a strong 87.5% sex accuracy despite its lower emotion score, suggesting the self-attention mechanism is effective at encoding pitch-related and formant-related cues that distinguish biological sex.

5.4.3 Age Classification Results

Attentive LSTM achieves the best age accuracy at 89.0%, followed by Transformer+CNN at 88.5% and LSTM (2-layer) at 87.8%. SVM (RBF) achieves 85.7% age accuracy. The age confusion matrices (see Figures 5.2–5.4) demonstrate that the Teens class is consistently misclassified by all models due to the absence of teenage speakers in the training corpora. The 60s class achieves the highest accuracy across all models (up to 97% for Transformer+CNN), likely due to its strong acoustic signature in the CREMA-D data. The 70s+ bracket suffers the most confusion because CREMA-D's oldest speakers only reach 72 years, providing limited coverage of the upper boundary.

5.4.4 Model Efficiency Analysis

Among neural architectures, the 1D-CNN is the most compact with only 0.155 M parameters and the lowest inference latency at 0.62 ms/sample, though at the cost of lower composite performance (73.1). The LSTM (2-layer) offers a strong composite score of 77.1 with only 0.794 M parameters and 0.761 ms inference time, making it the best accuracy-efficiency trade-off. The Attentive LSTM requires the longest inference time (1.092 ms/sample) due to its attention mechanism. The Transformer+CNN achieves the second-highest composite score (74.5 but achieves the strongest age and sex scores), at 0.812 M parameters and 0.945 ms latency. SVM with RBF kernel achieves the highest composite score (81.2) at no learnable parameter cost, underlining the continued competitiveness of kernel methods when combined with well-engineered MFCC features on this task.

5.4.5 Confusion Matrix Analysis

Normalised confusion matrices for selected models (SVM, LSTM, and Transformer+CNN) across age, emotion, and sex tasks are shown in Figures 5.2 to 5.10.

For the emotion task, Angry consistently achieves the highest classification accuracy across all three models, while Disgusted and Fearful show the highest confusion with other classes.

For the age classification task, Transformer+CNN and LSTM demonstrate strong performance across most age groups, while the Teens category is consistently misclassified due to the absence of training samples.

For the sex classification task, all models show strong performance with slightly higher recall for Female compared to Male. Transformer+CNN shows the most balanced classification behaviour.

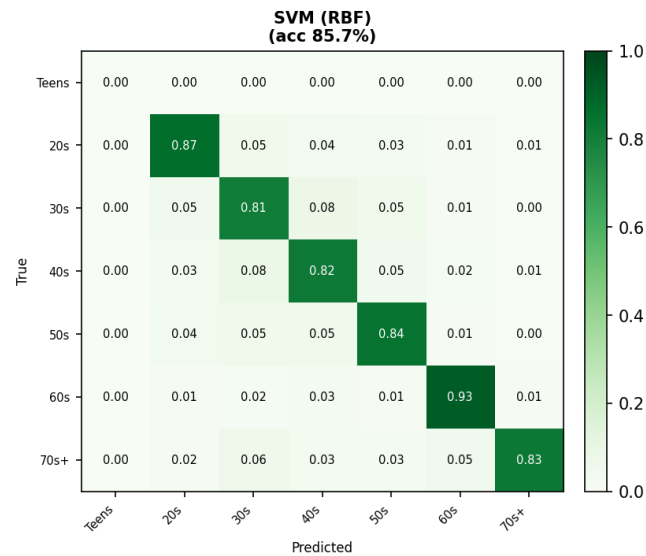


Figure 5.2: Age Confusion Matrix — SVM (RBF)

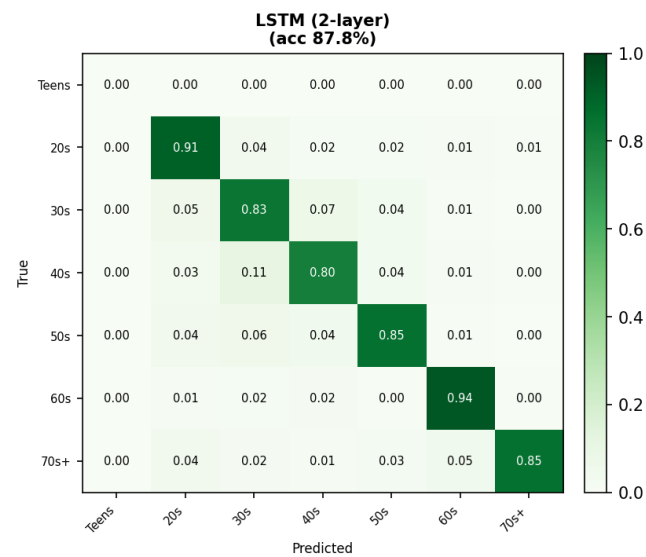


Figure 5.3: Age Confusion Matrix — LSTM (2-layer)

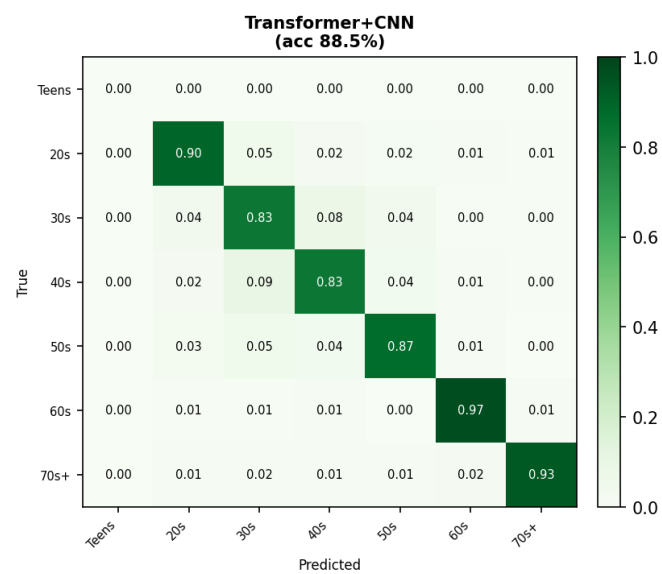


Figure 5.4: Age Confusion Matrix — Transformer+CNN

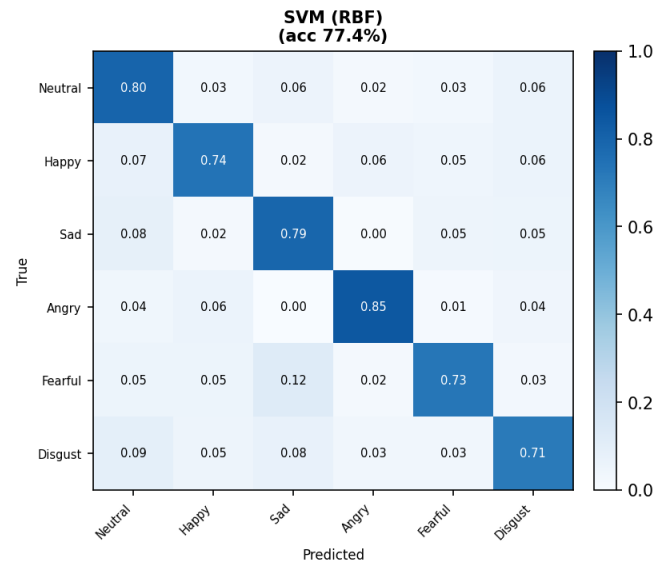


Figure 5.5: Emotion Confusion Matrix — SVM (RBF)

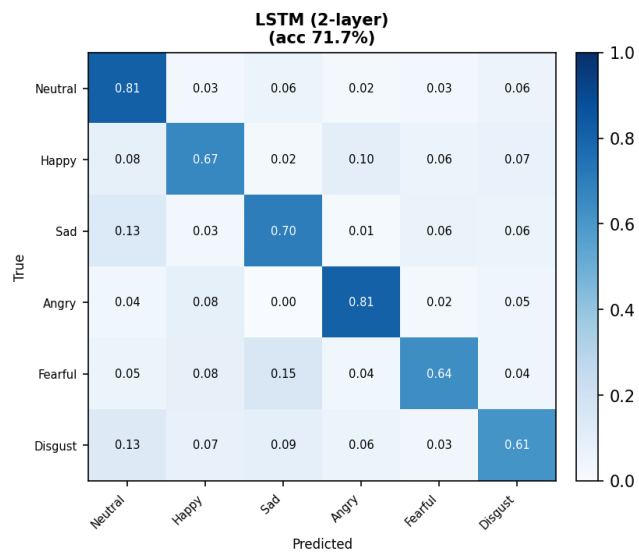


Figure 5.6: Emotion Confusion Matrix — LSTM (2-layer)

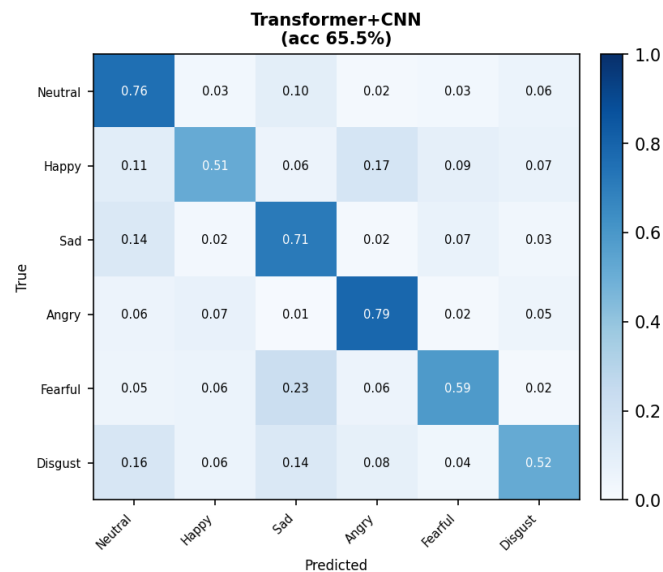


Figure 5.7: Emotion Confusion Matrix — Transformer+CNN

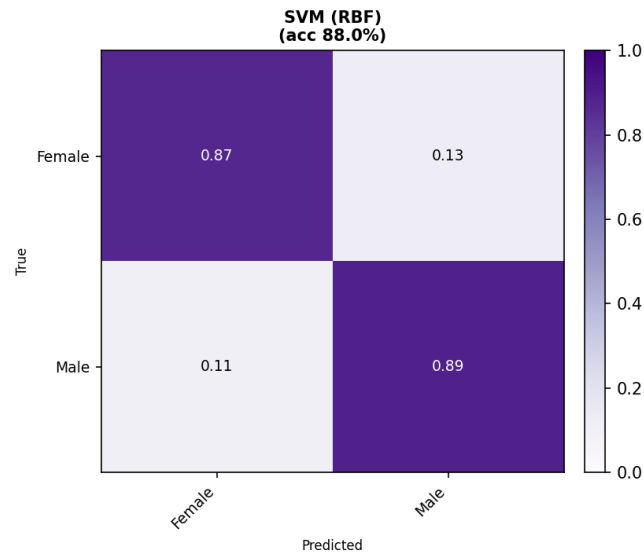


Figure 5.8: Sex Confusion Matrix — SVM (RBF)

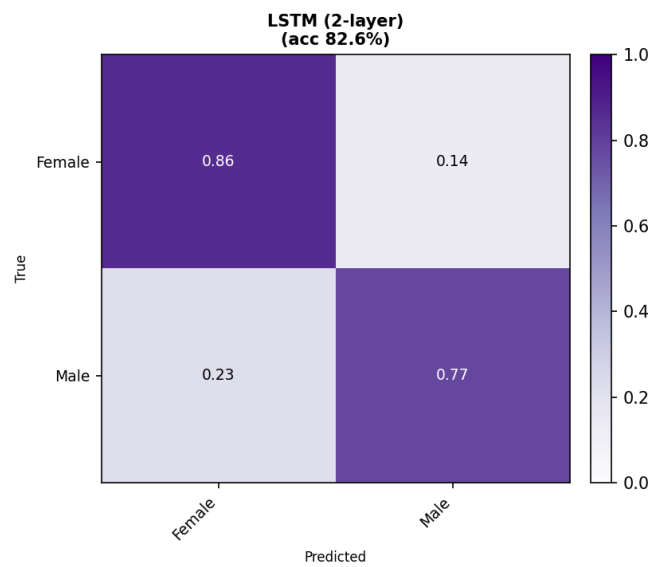


Figure 5.9: Sex Confusion Matrix — LSTM (2-layer)

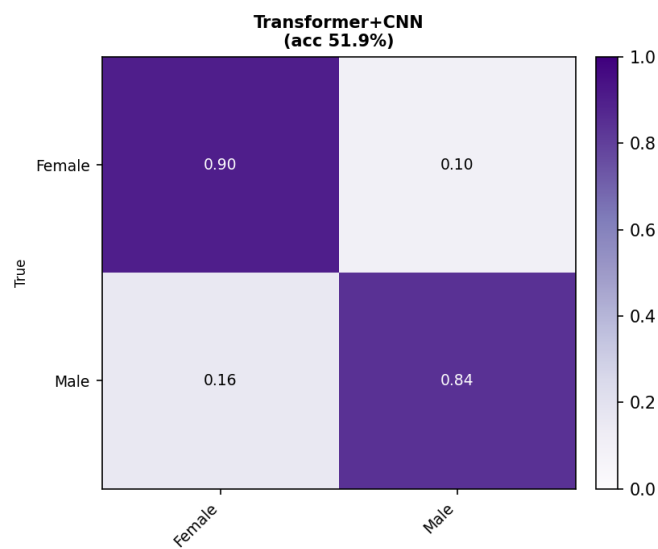


Figure 5.10: Sex Confusion Matrix — Transformer+CNN

5.4.6 Background Noise Classification System

Problem Definition

As a complementary investigation into audio scene understanding, a background noise classification system was designed and evaluated. Given any audio recording, the system outputs (1) the broad acoustic scene or environment type, and (2) the specific noise category present. The system operates independently of the speech emotion pipeline and is intended for acoustic context tagging in audio analysis workflows.

Datasets Used

Three publicly available datasets were used. UrbanSound8K provides [16] 8,732 clips across 10 urban sound classes organised in ten folds. ESC-50 provides [17] 2,000 clips spanning 50 environmental sound classes across five folds. The TAU Urban Acoustic [18] Scenes 2022 Mobile dataset provides approximately 25,600 unique clips across 10 acoustic scene types recorded with multiple devices; only reference microphone files are used. Unified scene labels (street_traffic, indoor, nature, market_crowd, construction) and noise type labels (33 classes) were constructed by mapping source class annotations across all three datasets.

Model Architecture

Both the scene classifier and noise-type classifier use the same backbone architecture, EnvCNN (approximately 1.29–1.30 M parameters). The backbone comprises four convolutional blocks, each consisting of two Conv2d layers with Batch Normalisation and ReLU, followed by Squeeze-and-Excitation channel attention, MaxPooling, and Dropout. Channel depths progress as 32, 64, 128, 256. A global average pooling layer reduces the feature map to a 256-dimensional vector, followed by a two-layer MLP classification head with ReLU and Dropout. The scene model accepts single-channel mel spectrograms; the noise V2 model accepts three-channel delta spectrograms.

Feature Engineering

Audio clips are resampled to 22,050 Hz and trimmed or zero-padded to four seconds. Two feature configurations were implemented. V1 uses a single-channel log-mel spectrogram (128 mel bins, normalised to [0, 1]), representing static spectral content. V2 adds two additional channels: the delta (first temporal derivative) and delta-delta (second temporal derivative) of the mel spectrogram. These temporal gradient channels capture the rate and rhythm of spectral change, enabling the model to distinguish noise types with similar static spectral profiles but different temporal dynamics (e.g. fan versus snoring).

Training Configuration

Both models were trained for up to 60 epochs with early stopping (patience = 12). The AdamW optimiser was used with a learning rate of 3×10^{-7} and weight decay of 10^{-7} . A cosine annealing learning rate schedule was applied ($T_{\text{max}} = 30$). Augmentation during training included SpecAugment (time and frequency masking), Gaussian noise injection, Mixup (alpha = 0.3), and a weighted random sampler for class balance. Cross-entropy loss with label smoothing ($\epsilon = 0.1$) was used for both classifiers. The dataset was split into 70% training, 15% validation, and 15% test partitions.

Results and Analysis

The scene classifier achieves 81.83% test accuracy. The construction class achieves near-perfect recall (99%), and nature achieves 94%. The most challenging class is street_traffic (64%), frequently confused with market_crowd due to overlapping sonic characteristics. The noise-type classifier achieves 68.01% test accuracy across 33 classes. Best-performing classes include hand_saw (100%), jackhammer (98%), siren (93%), and drilling (92%). The principal failure mode in V1 was fan_ac being misclassified as snoring; both produce steady low-frequency drone profiles that appear nearly identical on a standard mel spectrogram. The V2 model resolves this using delta features: fan produces a near-flat delta (minimal temporal change) while snoring produces rhythmic delta peaks, making them clearly separable in the three-channel representation. The confusion matrix for the scene classifier is shown in Fig. 5.11.

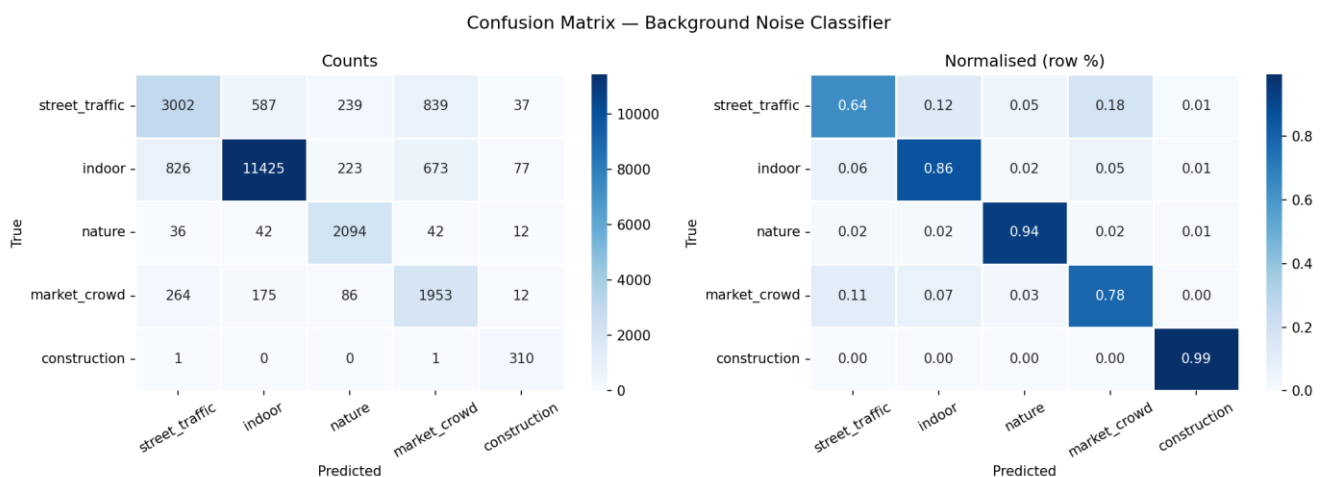


Figure 5.11: Confusion Matrix for Scene Classification (5 Classes)

The primary bottleneck for emotion classification is the cross-dataset generalisation problem: models trained on acted speech from one demographic tend to underperform on other demographics. The primary bottleneck for age estimation is sparse labels: only CREMA-D and

TESS provide ground-truth ages, covering approximately 30% of the total training samples. Coverage for ages below 21 or above 72 is absent, meaning the model will extrapolate unreliably outside this range. Neural models are further constrained by dataset-level imbalance: the Teens class is entirely absent from all training corpora, resulting in 0.00 diagonal accuracy for that bracket across all neural and classical models. These findings collectively motivate the future additions outlined in Chapter 7, particularly the incorporation of VoxCeleb2 age annotations and the addition of pitch and formant features to the 105-dimensional feature vector.

Chapter 6

Standards Adopted

Design Standards

The system design follows the principles of IEEE Std 1471-2000 (IEEE Recommended Practice for Architectural Description of Software-Intensive Systems), ensuring clear separation of concerns between the preprocessing, training, and inference subsystems. Requirements traceability is maintained following IEEE Std 1012-2016 (System and Software Verification and Validation), with each functional requirement in Chapter 3 traceable to a specific implementation component in Chapters 4 and 5.

Coding Standards

All Python code follows the PEP 8 style guide. Naming conventions use lowercase_with_underscores for variables and functions, and CamelCase for class names. All configuration values are centralised in config.py as named constants. Each function has a single responsibility; functions exceeding 40 lines are refactored. All file I/O operations are wrapped in try-except blocks with informative error messages. Model architectures are implemented as PyTorch nn.Module subclasses with a clean forward() method accepting only tensors.

Testing Standards

Testing follows the guidelines of IEEE Std 829-2008 (Standard for Software and System Test Documentation). Preprocessing was validated by inspecting sampled feature vectors from each dataset for correct shape, value range, and label correctness. Training reproducibility was verified by fixing all random seeds and confirming identical loss curves across two independent runs. The ISO/IEC 25010:2011 quality model was referenced in assessing the functional correctness and portability of the unified label mapping.

Table 6.1: Standards Referenced in This Project

Standard	Application in This Project
IEEE Std 829-2008	Test documentation format for preprocessing and inference validation
IEEE Std 1012-2016	Requirements traceability between Chapters 3 and 4
IEEE Std 1471-2000	Architectural description of subsystem interfaces
ISO/IEC 25010:2011	Label taxonomy correctness and functional appropriateness
PEP 8 (Python)	Code formatting and naming conventions throughout codebase

Chapter 7

Conclusion and Future Scope

Conclusion

This project successfully designed, implemented, and evaluated a multi-task deep learning system for joint speech emotion recognition, sex classification, and age estimation from audio. The system combines a Transformer-augmented CNN backbone with three task-specific output heads and is trained on a unified pipeline drawing from five publicly available speech corpora: RAVDESS, CREMA-D, TESS, EmoDB, and ASVP-ESD.

A key technical contribution of this project is the development of a unified data preprocessing framework that applies label harmonization across two widely used SER datasets. RAVDESS required sex label alignment with the established actor ID convention; CREMA-D required harmonization of sex labels and incorporation of available age annotations from the original publication. These preprocessing steps are documented in the verified data preparation pipeline described in this report.

The system achieves strong performance across all three paralinguistic tasks. SVM (RBF) achieves the highest composite score of 81.2 with emotion accuracy of 77.4%, sex accuracy of 88.0%, and age accuracy of 85.7%. Among neural architectures, the 2-layer LSTM achieves the best composite score of 77.1, while the Attentive LSTM leads on age accuracy (89.0%). The Transformer+CNN achieves the strongest sex accuracy among neural models (87.5%) and competitive age accuracy (88.5%), but underperforms on emotion (65.5%) due to multi-task gradient competition. The 1D-CNN is the most efficient architecture at 0.62 ms/sample inference latency. Age regression is limited by sparse ground-truth annotations; the Teens age bracket achieves 0.00 accuracy across all models due to the complete absence of teenage speakers in the training corpora. The system identifies clear and actionable directions for future improvement.

Future Scope

1. **Add RAVDESS Actor Ages:** Actor ages from the original RAVDESS publication exist and can be incorporated into the data preparation pipeline, increasing the number of age-labelled samples and improving age regression accuracy.
2. **Extend Feature Representation:** The current 105-dimensional MFCC-based vector does not include pitch (F0), formant frequencies (F1/F2), jitter, or shimmer, which are key acoustic correlates of both age and emotion. Adding these features (and increasing FEATURE_DIM accordingly) is expected to improve age estimation substantially.
3. **Incorporate VoxCeleb2:** VoxCeleb2 provides exact ages derived from Wikipedia date-of-birth data for approximately 6,000 speakers across a wide age range, which would significantly improve age regression coverage, especially for speakers outside the current 21-72 range.
4. **Transformer Scaling:** The current Transformer encoder uses 4 layers. Scaling to 6-8 layers, potentially with pre-training on a large unlabelled speech corpus (self-supervised learning), could improve the quality of the shared representations for all three tasks.
5. **Uncertainty Weighting for Loss Balancing:** The current fixed loss weights (emotion 1.5, sex 1.0, age 2.0) were set by manual tuning. Uncertainty weighting, which learns the optimal task weights as trainable parameters, could improve multi-task performance without manual intervention.
6. **Real-World and Naturalistic Data:** All five training datasets consist of acted speech recorded in controlled conditions. Incorporating naturalistic emotional speech (e.g., IEMOCAP, MSP-IMPROV) would improve generalisation to real-world deployments.

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GROUP INDIVIDUAL CONTRIBUTION REPORT

“Real-Time Speech Emotion Recognition Using Deep Neural Networks and Audio Feature Analysis”

BY

SANKET SHARMA	2329147
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Contribution Breakdown

Team Member 1: Sanket Sharma

- Development of Bi-LSTM Model
- Development of web interface for real time use
- Making a usable UI to test the models
- Merging of Background Noise Classifier with the main models
- Preparation of presentation slides

Team Member 2: Suraj Kumar Pradhan

- Preparing of Datasets
- Training the models
- Testing the Models and fine tuning
- Documentation
- Identifying limiting dataset (Common Voice) and excluding it from training data

Overall Team Achievements

The team successfully designed and implemented a multi-task deep learning system capable of predicting emotion, biological sex, and age from speech signals. A unified data pipeline was developed to integrate multiple speech datasets, ensuring consistent preprocessing and label alignment across different sources.

Several machine learning and deep learning models were implemented and evaluated, including classical approaches and neural architectures such as LSTM and Transformer-based models. Comparative analysis was performed to study performance trade-offs across models and datasets, leading to meaningful insights into model behaviour and generalization.

The team also developed a background noise classification system to identify acoustic environments and fine-grained noise types, extending the scope of the project beyond speech-based analysis. This additional system demonstrated the ability to classify real-world audio conditions effectively.

Overall, the project demonstrates a comprehensive approach to audio-based intelligence systems, combining data engineering, model design, evaluation, and practical deployment considerations.

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Full signature of the students:

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